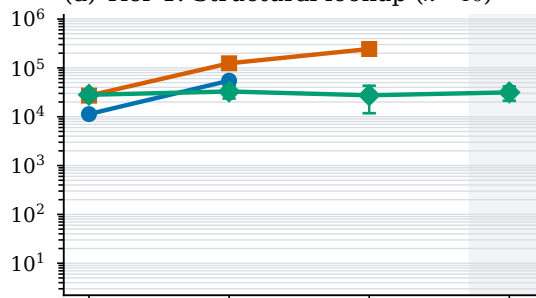


Mean token use across models

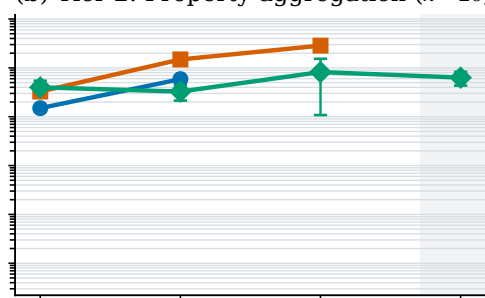
LLM CodeAct RLM

Recorded tokens per answered trajectory

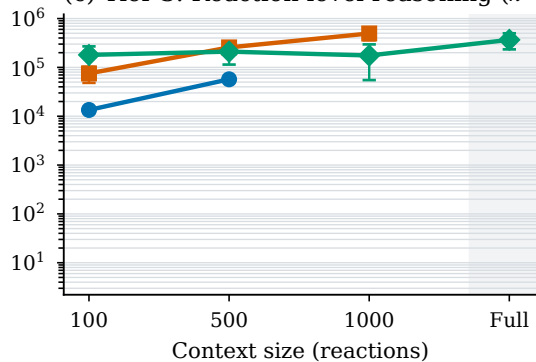
(a) Tier 1: Structural lookup ($n=10$)



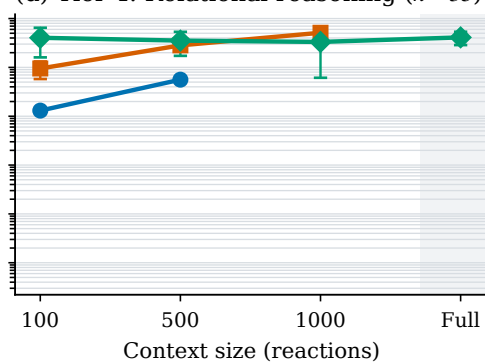
(b) Tier 2: Property aggregation ($n=20$)



(c) Tier 3: Reaction-level reasoning ($n=35$)



(d) Tier 4: Relational reasoning ($n=35$)



Unweighted mean $\pm$ SEM across terminal model arms. Accounting covers answered trajectories; failed-attempt token metadata are unavailable.